HASH BASED GRAYSCALE IMAGE INTEGRITY FRAMEWORK FOR TAMPER DETECTION AND RECOVERY

Report submitted to the SASTRA Deemed to be University in partial fulfilment of the requirements for the award of the degree of

B. Tech. Electronics and Communication Engineering

Submitted by

ATMAKURU LAKSHMI SAI SAHITHI
(Reg. No.: 127004033)
SAMADHI SATHVIKA
(Reg. No.: 127004223)

May-2026

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR, TAMIL NADU, INDIA 613 401

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR – 613 401

Bonafide Certificate

This is to certify that the thesis titled **“Hash based grayscale image integrity framework for tamper detection and Recovery”** submitted in partial fulfillment of the requirements for the award of the degree **of B.Tech Electronics and Communication Engineering** to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Ms.A.L.S.Sahithi(127004033)** and **Ms.S.Sathvika(127004223)** during the Sixth semester of the academic year 2025-26, in the **School of Electrical & Electronics Engineering**, under my supervision. This thesis has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University

Signature of Project Supervisor	:
Name with Affiliation	: **Dr. C. Nithya, SAP / SEEE, SASTRA Deemed University**
Date	: 01-05-2026

Project *Viva voce* held on ________________

Examiner 1 **Examiner 2**

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR – 613 401

Declaration

We declare that the thesis titled Project **"Hash based grayscale image integrity framework for tamper detection and Recovery"** submitted-by-us is an original work done by us under the guidance of **Dr.C.Nithya**, **SAP, School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the Sixth semester of the academic year **2025-26**, in the **School of Electrical and Electronics Engineering**. The work is original and wherever we have used materials from other sources, we have given due credit-and-cited them in the text of the thesis. This thesis-has-not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the candidate (1) : A·Sahithi

Name of the candidate (1) : **Atmakuru Lakshmi Sai Sahithi**

Signature of the candidate (2) : Sathvika

Name of the candidate (2) : **Samadhi Sathvika**

Date : 01-05-2026

Acknowledgement

We would like to express our sincere gratitude to our Honourable Chancellor **Prof. R. Sethuraman**, for providing us with the opportunity and the necessary infrastructure to carry out this project as part of our curriculum.

We extend our heartfelt thanks to our Honourable Vice-Chancellor, **Dr. S. Vaidhyasubramaniam**, and **Dr. S. Swaminathan**, Dean, Planning & Development, for their constant encouragement and strategic support throughout our academic journey.

We would also like to thank **Dr. R. Chandramouli**, Registrar for providing us with the opportunity to undertake this project.

We express our sincere appreciation to **Dr. K. Thenmozhi**, Dean, SEEE; **Dr. R. John Bosco Balaguru**, Dean (Sponsored Research); and **Dr. A. Krishnamoorthy**, Associate Dean (Academics), for introducing and facilitating this project as part of our curriculum.
We would like to convey our deep sense of gratitude to our **project guide**, Professor, School of Electrical & Electronics, for **Dr. C. Nithya** (SAP/SEEE)

We sincerely thank the P**roject Review Panel Members Dr. Siva J,** Associate Professor,
Mrs. M. Backia Lakshmi, Research Assistant for their valuable suggestions and constructive feedback throughout the course of this project. Their insightful observations greatly contributed to improving the quality and depth of our work.

We extend our appreciation to the all the teaching and non-teaching staff of the School of Electrical and Electronics for their support and assistance, directly or indirectly, in the completion of this project.

Finally, we gratefully acknowledge the encouragement and support of our family and friends, which played a vital role in the successful completion of this project.

ABSTRACT

In recent times, images are widely used for communication in critical domains such as medical imaging, surveillance, and forensic investigations. However, digital images are easy to tamper due to the availability of advanced editing tools .Unauthorized changes can affect the accuracy of data leading to major concerns in information security. Therefore, ensuring image authenticity and detecting alterations has become very important.This project is dedicated to the design of a secure framework for tamper detection and recovery in 256 × 256 grayscale images using cryptographic hashing techniques. The hashing technique used to generate the authentication codes for the image data is SHA-256.. Because even a minute difference in the data will result in a totally new hash code, this will aid us in determining which regions have been tampered with.Besides detecting the tampering, this system can also recover as much as possible of the corrupted parts of the image. This will avoid any degradation of the visual fidelity of the image. We will be able to recover the information contained in the image data without degrading the overall quality of the image.

Specific Contribution:

- Studied and implemented SHA-256 hashing algorithm for block-wise image authentication.
- Designed preprocessing stage including grayscale conversion and LSB clearing for consistent hash generation.
- Developed block division technique (8×8) for localized tamper detection.
- Implemented hash comparison logic for identifying tampered regions in the image.
- Contributed to overall system integration and testing of tamper detection module.

Specific Learning:

- Gained strong understanding of cryptographic hash functions and their role in data integrity.
- Learned image preprocessing techniques and grayscale image handling.
- Understood block-based processing for efficient image analysis.
- Gained practical knowledge in implementing security algorithms in image processing.

Signature of the Guide

Name: Dr. C. Nithya (SAP/SEEE)

Student Reg.No. : 127004033

Name: Atmakuru Lakshmi Sai Sahithi

ABSTRACT

In recent times, images are widely used for communication in critical domains such as medical imaging, surveillance, and forensic investigations. However, digital images are easy to tamper due to the availability of advanced editing tools .Unauthorized changes can affect the accuracy of data leading to major concerns in information security. Therefore, ensuring image authenticity and detecting alterations has become very important.This project is dedicated to the design of a secure framework for tamper detection and recovery in 256 × 256 grayscale images using cryptographic hashing techniques. The hashing technique used to generate the authentication codes for the image data is SHA-256.. Because even a minute difference in the data will result in a totally new hash code, this will aid us in determining which regions have been tampered with.Besides detecting the tampering, this system can also recover as much as possible of the corrupted parts of the image. This will avoid any degradation of the visual fidelity of the image. We will be able to recover the information contained in the image data without degrading the overall quality of the image.

Specific Contribution:

- Designed and implemented recovery data generation using MSB extraction and column-wise averaging.
- Developed LSB embedding technique to store hash and recovery data within the image.
- Implemented tamper recovery process using MSB restoration and LSB reconstruction.
- Performed performance evaluation using PSNR metrics.
- Assisted in testing, debugging, and validation of the complete system.

Specific Learning:

- Learned data hiding techniques such as Least Significant Bit (LSB) substitution.
- Gained understanding of image recovery methods and reconstruction techniques.
- Understood performance metrics like PSNR and their significance in image quality evaluation.

Signature of the Guide

Name: Dr. C. Nithya (SAP/SEEE)

Student Reg.No. : 127004223

Name: Samadhi Sathvika

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER-1

INTRODUCTION

1.1 Digital Images and Their Importance in Communication

Digital images are two-dimensional visual representations stored digitally as a matrix of pixels. Each pixel has an intensity value that relates to how light or dark the pixel is, and grayscale images use the 8-bit binary code to represent 256 levels of intensity, ranging from black through multiple shades of gray to white. Digital images may be taken using many types of devices, including cameras, scanners, and sensors, and they have been adapted into an array of different types of applications due to their ease of storage, processing, and transfer.

In modern communication systems, digital images serve the dual role of being both a means of efficiently transmitting information and serving as an effective medium for transmitting information in modern communications. In many cases, images can be used to represent information more clearly than text because they are visually equivalent to an actual object or visual representation of the information being transmitted. The development of the internet and wireless communication technologies has led to increased capabilities for the transfer of images, thereby expanding the range of possible applications for using images in modern communications. Some examples of common applications for using images include social media platforms, video conferencing, medical imaging applications, satellite communications, and surveillance and multimedia services.

Digital images have many advantages, but one of the major advantages is the ability to process and manipulate digital images in many different ways. This flexibility allows for enhanced, compressed and analysed in many forms and can be used in many applications, such as in the world of medicine with digital images of X-rays or MRI scans there are a lot moreoptions allowing doctors to determine how to appropriately diagnose disease. Proof! for example, the surveillance systems use image captured by the camera for monitoring or security purposes. In addition, an image is also used by satellites to help forecast the weather and to monitor the environment.

While the flexibility of digital images created all these different options creates more use-cases for them, this flexibility also creates exposure to the unauthorized modification of digital images.

The technology-enabled editing tools available today have made it easy for individuals to manipulate a digital image without leaving a visible mark after editing

In response to the increasing concern over the security and integrity of digital images, securing digital images as they are stored and transmitted has become an important area of research. In order to preserve the authenticity of images and identify tampered images, researchers continue developing techniques to provide an assurance of trust in digital communications systems.

1.2 Image Tampering and Real-World Examples

Tampering is when someone changes the content or meaning of a digital image by modifying the original. You can adjust the brightness/contrast, add/remove items from that image or anything else you find necessary.

There are several types of tampering techniques. One common one is to copy an area of an image and then paste it into the same image to conceal/hide or duplicate an object in question. Another example is splicing (also called composite images), where two or more separate images are used to create a new one. Retouching is another type of tampering; it typically involves making minor modifications and enhancements to aspects of the appearance of an image.

Tampering of images can pose many problems in various uses of these images. In the field of journalism/ media, for example, manipulating an image could result in the spread of incorrect information regarding the events captured in the images. If, for example, the image in a news piece about particular events were manipulated to alter the interpretation of those events, then there would be misinformation about the events captured in that picture, thus providing an inaccurate account.
Medical images, such as those from X-ray and MRI scans, could be altered, thus giving wrong diagnoses of the medical condition of a person and posing risks to patients' lives. In addition, image manipulation in security, including military and surveillance systems, may create wrong interpretations and decisions, compromising the country's national security. Image manipulation also is common in identity verification and fraud cases (fake IDs and documents).

Therefore, it becomes paramount to have methods of image tampering detection and where possible, recovery of original information. This means that image integrity is essential in the digital communication process.

1.3 Problem Statement

In today's digital age, images are extensively utilized for purposes ranging from communication and recording to making informed decisions. Nevertheless, due to the existence of advanced image manipulation software, it becomes easy to edit images in a manner that does not show any evidence of manipulation. This complicates the process of verifying the legitimacy of an image.Methods that have been developed to authenticate images are either concerned solely with tampering detection or lack an efficient strategy for restoring the edited areas. In fields including medical imagery, forensics, and military communications, it is insufficient to only detect tampering; in addition, it is equally important to reconstruct the image's content. Hence, there is a demand for developing a reliable system capable of detecting and recovering tampered areas in digital images.

1.4 Objectives

- Goals for the proposed project include:
- Detecting Tampering of Digital Images
    - Develop a reliable detection method based on hashed data to identify any alterations made on the perimeter of an image.
- Generate Recovery Information Per Image Block
    - Once a permanent image is completed, extract all pixel data related to the original image and store that information to allow reconstruction of tampered areas.
- Integrate Authentication and Recovery Information into the Image
- Use effective techniques to embed authentication and recovery information without degrading image quality.
- Using Embedded Recovery Data Along with Data From Other Image Blocks Reconstruct Any Tampered Area(s) of the Image
- Maintain Image Quality After Embedding and Recovery

- Ensure all images that were either embedded or recovered maintain a high level of visual quality by measuring them according to standardized performance metrics such as PSNR.

1.5 Applications

1. Medical Imaging

Medical experts diagnose the illness and treat patients with the help of digital records, including X-rays, CT scans and MRIs. Such manipulation of these records can lead to misdiagnosis and put the life of the patient at great risk. Thus, assuring the authenticity and integrity of these medical records is essential. The project provides a means of detecting modification and retrieves original information which enhances the reliability of medical decisions.

2. Military and Defense

Satellites, drones, and surveillance systems provide images for strategic planning and conduct of operations in defense and military missions. The manipulation of these images could lead to faulty intelligence analysis and/or potentially hazardous decision making. Tamper detection and image recovery techniques allow the user to verify the integrity of images received and provide assurance to the user that the images are original and authentic. Image tamper detection and recovery techniques improve the nation’s security by increasing the confidence of users in the ability to collect accurate intelligence.

3. Digital Forensics

Digital forensic professionals frequently rely on photographic images to provide evidence for law enforcement investigation and prosecution as well as civil lawsuits. If an image that has been captured is modified in any way, the results of the investigation and legal proceeding could be altered negatively, thus diminishing the weight of the evidence associated with that image. The developed solution enables forensic experts to check whether or not the photographic image has been manipulated. The restoration of the original image in the case where the image has been manipulated. This will help to ensure the reliability and authenticity of digital forensic evidence.

3. Journalism and Media

Image plays a significant role in the field of journalism and media as far as presenting news is concerned. On the other hand, manipulated or altered images may result in the dissemination of incorrect news leading to misinformation of the audience. The manipulation of a news image can change the narrative of the situation entirely. In this case the authenticity of the image needs to be confirmed any manipulation of the image can be detected and the original content can be recovered in the proposed solution.

CHAPTER-2

RELATED WORKS

2.1 Literature Survey

The paper [1] proposes a triple self-embedding fragile watermarking method for detecting and recovering tampered image regions. The image is divided into small blocks, and for each block, both authentication and recovery information are generated. The authentication bits are used to detect tampering, while recovery bits help reconstruct modified regions. A key feature of this method is that recovery data is embedded into multiple (three) different blocks, improving robustness against large-area tampering. During detection, extracted authentication bits are compared with recalculated values to identify tampered blocks. In the recovery phase, data from untampered partner blocks is used to reconstruct the image. The method shows improved recovery performance and better image quality compared to traditional approaches. However, recovery may fail if all corresponding partner blocks are tampered.

The paper [2] presents an image integrity verification system using a combination of perceptual hashing and watermarking techniques. The approach adopts an FJLT hashing technique for generating a reliable hash code for the image. This hash code will be subsequently integrated into the image via a blind watermarking scheme based on DCT.. At the receiver, the hash is extracted and compared with the newly generated hash to verify image authenticity. This technique is meant to be tolerant of acceptable alterations like JPEG encoding but able to detect any malicious tampering. According to tests conducted, this technique is successful in distinguishing between benign and malicious alterations.. Experimental results show that the method can effectively distinguish between normal compression and intentional tampering.

The paper [3] proposes a fragile watermarking method for image integrity verification using the SHA-256 hash function is proposed in the paper. The image is divided into non-overlapping 32×32 blocks, which are further split into four 16×16 sub-blocks. A 256-bit hash value is computed from three sub-blocks and used as a watermark. The watermark is embedded into the fourth sub-block using a least significant bit (LSB) based blind watermarking scheme . On the receiver, the watermark is extracted and compared with a newly computed hash to detect tampering. The method enables accurate localization of the modified regions in the image. This differs from existing techniques by using the entire hash value and thus making it more reliable and secure. Experimental results demonstrate that the proposed method can achieve high visual quality and detect all types of image tampering efficiently.

The paper [4] proposes a novel efficient dual watermarking scheme for detecting and recovering from any tampering attacks in images. In the proposed scheme, the image is divided into non-overlapping 2 × 2 blocks and watermark information is extracted based on the characteristics of the blocks, like average pixel intensity and parity bits. Then, two copies of the watermark information are inserted into the blocks using the least significant bit replacement strategy. To detect tampered areas in the image, a hierarchical detection framework consisting of three layers is proposed. Similarly, to recover the tampered areas in the image, a two-layer reconstruction approach is utilized. An enhanced smoothing process is introduced to minimize distortion. Experimentation demonstrates that the proposed scheme is capable of detecting and restoring the tampered image to its original form with PSNR ranging between 43-45 dB.

The paper [5] proposes a fragile watermarking technique for image authentication and recovery using a combination of Sudoku puzzle mapping and the MD5 hash algorithm. The method generates authentication data using hash values and embeds recovery information within the image itself using a self-embedding approach. Sudoku-based block mapping is used to securely distribute watermark data across the image, improving resistance to tampering. During detection, the extracted hash is compared with recalculated values to accurately identify tampered regions. The recovery process reconstructs altered areas using the embedded information. The technique achieves high accuracy in tamper localization and maintains good visual quality. However, performance may degrade when large portions of the image are tampered.

The paper [6] proposes a reversible image authentication (RIA) scheme for accurate tamper detection and block-wise localization. The method uses reversible watermarking and prediction-error histogram shifting (PEHS) to embed authentication and recovery information within image blocks. It improves the number of embeddable blocks while reducing image distortion compared to existing techniques. Authentication codes are generated using block features such as pixel values, coordinates, and neighboring information along with a cryptographic hash function. The system effectively detects tampering caused by splicing, copy-move, and collage attacks with high localization accuracy. Experimental results show improved detection performance and efficiency suitable for real-time and resource-constrained applications. However, performance depends on embedding capacity and block characteristics.

The paper [7] proposes an efficient dual watermarking method for image tamper detection and recovery. The method divides the image into non-overlapping 2×2 blocks and generates watermark information by using block features such as average intensity and parity bits. The watermark data are duplicated and embedded in different blocks through LSB substitution, providing redundancy for reliable recovery. The tampered regions are detected by a three-level hierarchical detection mechanism and the altered parts of the image are reconstructed by a two-stage recovery process. The scheme also involves a smoothing function to reduce distortion and maintain image quality. The experimental results show that the proposed method can effectively detect the tampering and recover the original image with good visual quality, in which the PSNR values are about 43-45 dB

The paper [8] presents a hash-based fragile watermarking technique for tamper detection and localization of digital images. The method divides the image into 4×4 non-overlapping blocks and applies Discrete Cosine Transform (DCT) to extract DC coefficients. A SHA-256 hash is generated from these coefficients, and a portion of the hash is used as watermark data. The image is then scrambled using Arnold transform, and the watermark is embedded into the blocks using LSB substitution. At the receiver, the watermark is extracted and compared with a newly generated hash to detect and localize tampering. The technique is robust against various attacks such as noise, compression, rotation, copy-move, and copy-paste operations. Experimental results show that the method achieves high image quality with an average PSNR of about 51 dB and accurately detects and localizes tampered regions.

The paper [9] conducts an extensive review of the area of medical image authentication, tampering detection, localization, and tampered region recovery. This work emphasizes the need for the protection of integrity and authenticity of the medical images since any tampering may cause diagnostic and security problems. First of all, various medical images formats and medical imaging techniques are discussed, and their importance in terms of protection of ROI (Region of Interest) and RONI (Region of Non-Interest) is addressed. Three types of existing techniques are identified, including watermarking methods, digital signature methods, and hybrid ones, which unite both approaches mentioned above. As for the comparative analysis, it is carried out in the following fields: robustness, tamper detection, localization, tamper recovery. Thus, it can be stated that even though there exist many effective approaches to authentication and tamper detection, localization and recovery pose significant problems.

The paper [10] proposes a hybrid-Sudoku watermarking method for digital tampering detection and localization. The hybrid-Sudoku-based approach utilizes a double-perspective Sudoku-based data hiding technique, enhancing the security and robustness of the system. Watermarking in this case involves generation of watermarks using the pseudo-random number generator and embedding of the watermark into pairs of pixels of the image via a two-layer embedding process. In order to extract the watermark in the receiver's end and check its authenticity, a two-tiered extraction method is employed. Localization occurs at the level of individual pixel pairs and a two-layer tampering detection mechanism is employed to enhance accuracy. From experiments carried out on the algorithm, it has been established that the method can detect tampering at very high accuracy rates (with TDR of up to 99.72%). Moreover, it preserves the integrity of the image with PSNR of approximately 47.8 dB .

2.2 Existing System

There are various methods to validate the integrity of digital pictures by detecting alterations in those digital images. Most approaches are based on identifying the presence of alterations in the image but suffer from limitations regarding the ability to recover the altered area efficiently based on the alterations.

One common method for validating a digital image is to use a digital watermark embedded into the digital image (or possibly in the image and the watermark). When the image is altered, the digital watermark is affected, which would therefore indicate that the digital image has been tampered with. Watermarking is a good method for detecting alterations to a digital image; however, because of how it is applied/embedded into the image, it tends to also cause slight distortion in the digital image, and it may not always allow for exact recovery of the altered regions of the digital image.

Another method used frequently in digital image validation is hash-based validation, where a hash value is created based on the entire digital image or based on blocks of the digital image. This means that any changes in the digital image will cause hashing alterations, which will create a different hash value. So, hashing is highly effective for detecting tampered images; however, the current methods that use hashing for non-recovery purposes only provide for detecting tampered images without the ability to use hashing to recover the altered area of the digital image.

Block techniques have been previously adopted in systems in which images were divided into blocks, and at least one of those blocks was processed by the current system, enabling the more accurate detection of localized tampered areas than more traditional approaches. In many situations, however, there will be some recovery metadata that has been stored within the same block, which makes all of that information susceptible to loss if a block is completely tampered with. Therefore, if you have a completely tampered block, all recovery information associated with that block will also be lost and thus hinder any attempt to reconstruct or restore that image.

In addition, the existing systems that are currently available tend to be complicated and require a significant amount of storage in order to store extra data. They are also generally very susceptible to noise and the effects of compression, which makes them unsuitable as an effective solution for use in real-world applications.

2.3 Limitations of Existing System

- Loss of recovery data when blocks are fully tampered
- Image quality degradation due to embedding
- High computational complexity
- Limited robustness against noise and attacks

2.4 Need for Proposed System

There need to be developed systems capable of detecting tampering and providing an efficient means of recovering altered regions of the image while retaining high quality and robustness, as a result of the above mentioned limitations.

CHAPTER-3

METHODOLOGY

3.1 Proposed System

To address current shortcomings, a highly effective and robust technique has been proposed for detecting and recovering from digital image tampering issues. By using hash-based authentication combined with block-based recovery techniques, both accuracy and reliability can be ensured through this proposed approach.

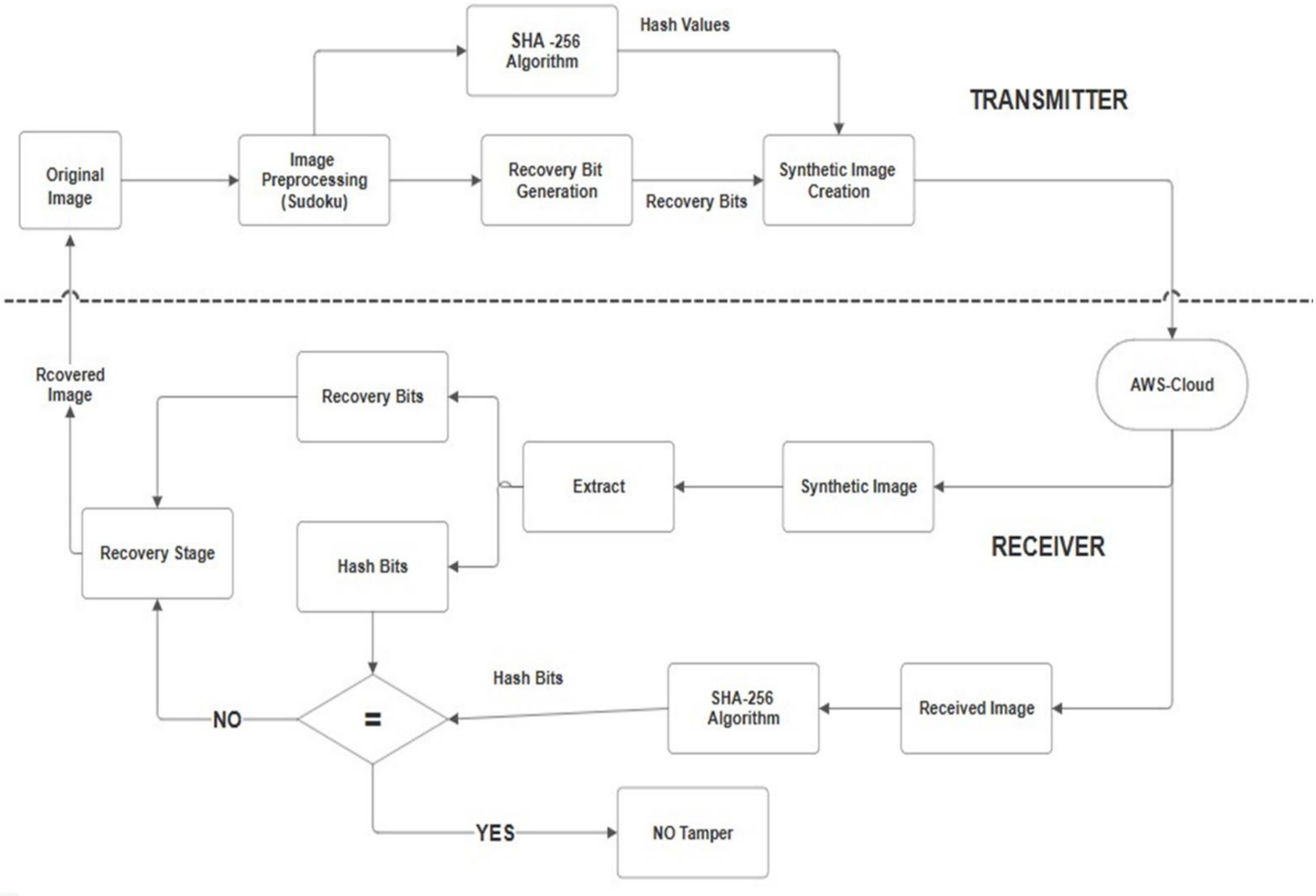


Fig 3.1 Block Diagram of Proposed Image Tamper Detection and Recovery System

Figure 3.1 shows the architecture of the proposed system for detecting and recovering an image from tampering. The original image is subjected to preprocessing by means of Sudoku, hashing through SHA-256 algorithm, and embedding of recovery bits to get a synthetic image. The received hash bit values are compared against freshly calculated hash values at the receiver side, and recovery bits are utilized to reconstruct tampered images.

The grayscale input image is initially resized to fit into smaller blocks of 8×8 pixels before being processed independently, thus yielding more precise information about tampered areas of the image. For example, if a block undergoes any alteration (e.g., changes) at this stage (e.g., changes to pixel values), then it will generate an entirely different hash value (e.g., SHA-256) as opposed to its pre-altered state. Therefore, if a tampered area exists based on this method, then a corresponding hash may also exist.The organization generates an additional data set for each image block based on the hash technique to assist with recovery in case of tampering (e.g., corruption). This will provide enough data to rebuild or reconstruct a block in the event of a corruption incident. The recovery dataset is created from extracting the most significant bits (MSBs) of each pixel value and creating a 32-bit representation for each pixel value as a reduced format.The generated hash values and recovery data are then embedded into the image using a suitable embedding technique, such as least significant bit (LSB) substitution. The resulting image is called the synthetic image, which is transmitted through a communication channel, such as cloud storage (AWS). At the receiver's end, the image will once again be broken down into blocks and the hash values recomputed and then compared to the extracted hash values to identify any tampering. If there is a mismatch, the block is flagged as tampered.To recover data, an embedded recovery mechanism and cross-block information will be used to recover that data. For example, the high-order bits of the block will be recovered using the transmitted data and the lower-order bits will be reconstructed from average values taken from other blocks that surround that block. Therefore, even if some of the blocks have been totally altered, some of that information can still be recovered by reference to other bloc

3.2 Image Preprocessing

Image preprocessing is the foundational and primary process for implementing the proposed algorithm. The central goal of this stage is to convert the input image to an appropriate form for further analysis (e.g., using hashing or embedding). Correctly performing this preprocessing stage ensures that the final result will be accurate and verifiable throughout the evidential investigation of tampering and recovery. The original image will initially be converted to grayscale (i.e., each pixel will only have one intensity represented between 0 and 255 by using either 8 bits or 256 levels). Thus, in grayscale, pixels can have 0 (black) to 255 (white) values. The conversion to grayscale will reduce the complexity of processing compared to having three color channels. Once the image has been converted to grayscale, the next operation to be performed is to set the LSB of each pixel to a value of 0, which can be done through the application of a bitwise AND operation, such as to 254 (binary = 11111110). As a result, this operation will clear the LSB of each pixel. Setting the least significant byte (LSB) to 0 guarantees that any information that has been previously embedded/hiding from an image does not affect how hashes are produced when the hash algorithm is extremely sensitive to changes, by removing the LSB from the computed hash's data will assure that all hashes produced with that data will produce the same result, and thus will protect against the false identification of alteration and guarantee space available for future embedding of new data; doing this helps standardise the images prior to any other process being performed on them and also allows for greater accuracy in determining whether a particular image has been tampered with as well as greater recovery of images that have been tampered with.

3.3 Block Division

Block division comes immediately after preprocessing, whereby the grayscale image is segmented into non-overlapping smaller sized blocks. Block division has a significant contribution to tamper detection and restoration of the images.

Step 1: Division into 8 x 8 Blocks
The grayscale image is divided into non-overlapping 8 x 8 blocks, each comprising of 8 pixel rows by 8 pixels columns making up 64 pixels per block.

Step 2: Calculation of Number of Blocks
For the example above, consider an image that has a dimension of 256 pixels by 256 pixels:

Step 3: Independent Processing of Blocks
Processing of each block independently allows the SHA-256 hash code and recovery-bit to be computed from them. Processing of blocks independently enables the system to detect tampering at a particular region of the image.

As observed, block division significantly improves the efficiency and accuracy of the proposed system**.**
Consider an image of size 256 × 256

Total number of blocks = (256 / 8) × (256 / 8)= 32 × 32= 1024 blocks
Each block:
Size = 8 × 8 pixels
Total pixels = 64 per block

3.4 SHA-256 Hash Generation

In the proposed system, creating a hash is critical to determine if tampering has occurred on an image. A hash function takes an input (data) and creates a fixed-length output (hash value or digest) through a mathematical process. Each time the hash function is applied to the same data, it results in the same hash value. However, even the slightest change to that data will result in an entirely different hash value. SHA-256 hashing algorithm is being used. SHA-256 is an industry-standard cryptographic hashing algorithm that generates a 256-bit (32-byte) hash value. It belongs to the SHA-2 family and is known for its high level of security and reliability. In this method, each 8×8 image block is converted into binary data and the SHA-256 algorithm is applied to create a unique hash value representing the block. The hash value being created will act as a digital fingerprint for that block within the image. When at least one pixel is altered within the block, the hash value will also change drastically as a result of the avalanche effect.

Cryptographic hash functions exhibit the avalanche effect. A small change made to an input will consequently lead to a significant and completely different output hash. The SHA-256 hash is therefore considered an extremely effective method for determining whether an input has been altered or not. SHA-256 is also considered an irreversible hash function. Once the input data is transformed into the SHA-256 hash, it is impossible (mathematically) to revert the input data from the hash.

In the proposed system, SHA-256 fulfils several purposes:

- Hashing the 8×8 blocks results in the generation of unique hashes for each block.
- The hash values from the 8×8 blocks are used to determine whether any portion of the input has been tampered with or not.
- The use of SHA-256 hashes ensures the integrity and security of the data.

3.5 Working of SHA-256 Algorithm

SHA-256 is a cryptographic hash function that generates a fixed-length 256-bit hash value from input data according to specified processes, providing both a high level of security and sensitivity to minor input changes. Thus, SHA-256 is effective for detecting if a digital image has been altered.

SHA-256 works as follows:

Step 1: Read the 8 x 8 block image as a input

The input to SHA-256 occurs in a binary format, where each 8 x 8 block of the image is converted to binary format since each pixel is represented by an 8-bit number prior to processing.

Step 2: Pad the Data

- The message is padded to ensure that its final length will be a multiple of 512 bits. Padding is created as follows:
- Add 1 (binary) to the end of the original message
- Add 0s until there are enough digits to bring the length to 512
- Add 64 bits which indicates the size of the original message
- The result will be a padded message of length that is a multiple of 512 bits.

Step 3: Message Division

The padded message is broken into many blocks of 512 bits. Each of those blocks will go through many calculations, and each block is independent of all the other blocks.

Step 4: Initialize Hash Values

SHA-256's hash is built using eight initial hash values (H0 to H7). They will each be the value assigned to the square root from prime numbers. These hash values will be used to initialize the hash computation process.

Step 5: Create the Message Schedule

The first 512 bits of the padded message are broken down further and turned into 16 words that are 32 bits long. After that, we expand each of these 32-bit words into 64 words (32 bits) using various logical and bitwise operations to create the message schedule.

Step 6: Compression Function (64 Round Processing)

- The core algorithm of SHA-256 has 64 rounds of processing, and in each round, we perform the following activities:
- Perform logical operations on the incoming data (such as AND, OR, XOR)

- Create the next hash (update temporary hash variables)
- Rotate the data or shift the data to the left by a specified amount
- Use different constants for each of the 64 rounds
- These processes create significant randomness and security in the data returned as output.

Step 7: Final Hash Computation

The final output of the proposed system is achieved by concatenating all 8 updated hash values (H0 through H7) together after they have been completed through their 64 rounds of hashing; creating a final 256-bit hash value from the hash values.

This hashing process has been put into place to:

- Ensure high sensitivity to tampering;
- Produce a unique hash for each block;
- Provide verification of security and integrity in the proposed system.

3.6 Recovery Data Generation

The proposed system also creates a method for recovering each 8×8 image block, in addition to creating a hash. This data is critical when trying to reconstruct an image that has had part of it altered before being transmitted from one user to another. Each eight-by-eight image block contains 64 pixels. Each pixel is 8 bits in size; therefore, if you were to store all of the pixel values for the 64-pixel block, the amount of data required would be quite large. Therefore, the proposed system uses a data reduction strategy to create a compact image block from the incoming data.

In general terms, the recovery data is created through the following steps:

Step 1: The 8×8 image block is laid out in a matrix format with 8 columns and 8 rows.

Step 2: Extract the 4 most significant bits (MSB) of each pixel from this matrix.

Step 3: After the MSB values have been extracted from each image block.

Step 4: The average of the MSB value for each column will represent the average intensity for that column.

Step 5: The computed average is then converted into a 4-bit value, representing the column information. Since there are 8 columns in a block, this process generates 8 average values, each consisting of 4 bits. Thus, the total recovery data for one block becomes,

Number of columns = 8

Bits per column = 4

Total recovery data = 8 × 4 = 32 bits

Step 6: This 32-bit representation efficiently captures the essential information of the block while significantly reducing the amount of data required.

```
Block (0,0) : 0d1cab444987ff7372cfcfcf63f785e6d379943efcd8dd2d422ca6c0e6005383 01101011101010101010101010101010
Block (0,1) : b0e15e47818f5e8022325de0671a9c2f2d85ee022c2f42f45e9ddb9a1fc04620 10101010101110111011101110111010
Block (0,2) : de7368433ffc8a277a82a914f4e160dc0a7e8e148ac25d992a0681bb99203157 10111011101110111011101110111011
Block (0,3) : d5cdcf89b2180d5e15588aa7a4133b320b6a4c06ffbeb80d1dbcabd8a4a62de5 10111011101110111011101110111011
Block (0,4) : 552a413eea91516bef59a15a4b7c755bd7f6831d28a0c8eaade84430f5990950 10111011101110111011101110111011
Block (0,5) : 4b50822ddb04a8e912842518d99e1f84d52d90548613684e63397b91b51f0939 10111011101110111011101110111011
Block (0,6) : 86e74ab4b479bf65b3d4c9d9ed595677ff2a261b34c13674394dabafc2a7fd6e 10111011101110111011101110111011
Block (0,7) : e279e4a7b1cf9635d38f3adfb9e3d9f40bedcab652f1ba6cb1e4886190ba6342 10111011101110111011101110111011
Block (0,8) : b5fd38f449bfe680a5f554fba0efdfebca58839287aba2c019f96c0131a7cb55 11001100101110111011110010111011
Block (0,9) : 5a8e6d98f63ef82194383eaa90169d6394a6edb02ff101511c66ab013025b3fb 10111011101111001011101110111011
Block (0,10) : b9d785f994cc264ecd9f5b896e7bfbbbd592ecf8e5968afef1884d710602389a 10111011101110111011101110111011
Block (0,11) : 4bb69535505a787d81457e550f9113d423adb2201c6cb76788f721697b9f3ec9 10111011101110111011101110111011
Block (0,12) : df53e74e24f5343ba255294133e248a96a118b89e1cfa5b58f3134120d9986bc 10111011101110111011101110111011
Block (0,13) : fc427c8b9518ce6ed85a5b8e7d97acbb0167a62e05a97035a5caa35dcc2ed4cc 10111011101110111011101110111011
Block (0,14) : e3760886708b9a5dff845bb3f5c711bdeb9e67b3295f518f427510a00ac7b45c 10111011101110111011101110111011
Block (0,15) : 79ced8e4a38ea134edb72c59fff757823a8832d2a6943479572293da18249883 10111011101110101011101010101010
Block (0,16) : 21186527adc52ebe028419b83fe5a838c6f819a65f2ef1d115e031ef21c50d50 10101010101010101001100110011001
Block (0,17) : 9b6cf6c4ed0d9181d5054c4f6d4c3d382b8b71cce68bca1f9f3616a60d481fdb 10101010101010101010101010101010
Block (0,18) : 3e9b3ac839e55a0c53d5c2c961b7728cbffcfe7d9d61fa3a901adc171b73d627 10101010101010101010101010101010
Block (0,19) : 9de3cff8670145c33a9f1acf1c6170b2687f8e1c41c6af90ee95298a04d3e9b7 10101010101010101010101010101010
Block (0,20) : 6f3e03338604b3a5eb7be24b1d92482a7a80f4a1bdb0aff6acd3b50763e14e27 10101010101010101010101010101010
Block (0,21) : 603d35cc58219611e893baa7194c80cd15f692290f2382e4e39b37040a772817 10101010100110011001100110011001
Block (0,22) : ab0769b76e6a294b8c69ddb01514f55c02467ea4c37db470b0f1159befbb92a1 10011000100010001000100010001000
Block (0,23) : 3179f76499f2681f4ec89c6d2a256594a3c5c71eca7042b89e52a60bbb4201ae 10001000100010000111011101110111
Block (0,24) : 0daad39321447d057eaa7a50bf42fd19beb4e15b01c02640542b6f4322eff050 01110111011101110111011101110111
Block (0,25) : e42adbfcc6d10722607110fae4d5a973522220f262cef839b884143ae14c719a 01110111011101100111011101110111
Block (0,26) : beb17cbd78dbedf68f30d6d43a905b727ea796f6f96cd20273348cb5e065430b 01110111011101110111011101110111
Block (0,27) : 3151901b8ae94ca3017c2855e1b328c1822e6ffa6a264c88e4c3564535eafd01 01110111011101110111011101110111
Block (0,28) : 188a6bee11dc2ebfb5090c7c33a6c54de76092e6c8418418656375240 1c4ed30 10001000100001111000100010001000
Block (0,29) : 66e31e1b2323e0bf5d2e6e369fe53a22a784ba197d787e6902b279e202d9a300 10011001101010101010101010101010
Block (0,30) : 7f318d57b1062b1fd7a563a75c02f14fc623b365bca5bba2048400d0a75ffdfb 10101001100110101010101010101010
Block (0,31) : 6aa02e0ede42a1e3070913949a49b8954d609ba80a89d2c8c29346186c057e2f 10101011101010101010101010101010
Block (1,0) : 92124af2ff9927a1d05d105cfc01dd1498ea889ffefeb7110f6a48c53ef40f6a 01011010101010101001100110101010
```

Fig 3.2 Block-wise Hash and Recovery Bits

Hash values generated along with the recovery bit information associated with each 8×8 image block are illustrated in Figure 3.2. These blocks are identified using the hash values generated for each image block along with recovery bit information for detecting tampering and image reconstruction purposes.

ORIGINAL IMAGE **SYNTHETIC IMAGE**

Fig 3.3 Original Image and its corresponding Synthetic image of Hash and Recovery Bits with the size of 192×192

Figure 3.3 Illustrates the synthetic image generated after embedding hash values using a 32-bit pixel representation. The image appears noise-like due to the embedding process, which distributes authentication and recovery information across pixel values. This transformation ensures that the embedded data remains imperceptible in structure while enabling effective tamper detection and recovery at the receiver side.

3.7 Example of Recovery Data Generation

Consider an example of an 8×8 image block. Each element in the block represents a pixel value ranging from 0 to 255.

Step 1: Consider an 8×8 Block

120	135	142	150	160	170	180	190
110	125	138	148	145	167	169	32
123	124	125	24	90	40	50	134
156	189	198	156	134	23	46	84
190	188	100	101	108	156	162	148
111	148	134	123	134	156	178	199
20	55	66	77	88	99	22	73
154	143	137	174	129	148	191	131

Step 2: Extract MSB (First 4 Bits)

Each pixel is an 8-bit value. We extract the first 4 bits (MSB) of each pixel.

Example:

120 → 01111000 → MSB = 0111

135 → 10000111 → MSB = 1000

This process is applied to all pixels in the block.

Step 3: Column-wise Processing

Now, each column is processed separately.

Example (Column 1):

Values: 120, 110, 100, 90, 80, 70, 60, 50

Extract MSB values → convert to decimal

Compute average of these values

Suppose the average value is 7, which in binary is 0111

Step 4: Repeat for All Columns. So, column-wise average values and its upper nibble values are tabulated in Table 3.1.

Table 3.1. Upper Nibble of Column-Wise Average Values

Column	Average (Decimal)	4-bit Representation
1	7	0111
2	8	1000
3	8	1000
4	9	1001
5	9	1001
6	10	1010
7	10	1010
8	11	1011

Step 5: Form 32-bit Recovery Data

Concatenate all 4-bit values:

0111 1000 1000 1001 1001 1010 1010 1011

This results in a 32 bit recovery sequence

3.8 Embedding Process

After generating the hash and recovery data for each 8x8 block, the next step in the proposed system is to embed the authentication and recovery information into the image itself, creating a secure and self-contained image known as the synthetic image. The embedding process is done using a method called Least Significant Bit (LSB) substitution. In LSB substitution, the least significant bits of the pixel values are replaced with the generated hash bits and recovery data bits. As the least significant bits contribute very little to the overall visual quality of an image, modifying them does not create any noticeable distortion.

For each block, the system embedded by two ways:

01. Hash values (used for tamper detection)
02. Recovery data (used to reconstruct the image)

The embedding points for these bits are located in the LSBs of each pixel value in the image. The embedding process has been carefully structured so that the receiver can correctly extract the data.

A synthetic image, which appears almost identical to the original image, will have some hidden data that is needed to detect tampering and obtain a recovery of the original image.

3.9 Tamper Detection

Tamper detection is a critical phase and takes place after each received synthetic image has been delivered successfully by the cloud. The purpose of this stage is to check if any of the information contained within the provided image has been altered either during transport or storage. The proposed system has included a hash based tamper detection methodology, represented by each of the eight-by-eight blocks of image pixels will have a hash value based on SHA-256 created for reference.

The tamper detection process is carried out as follows:

Step 1: A received image is divided into 8 x 8 blocks much like the image was divided on the sending side.

Step 2: An embedded hash value for each block is extracted.

Step 3: The system will recomputed the hash for the same block by using the current pixel values of the block.

Step 4: The embedded and recomputed hash value are then compared.

1. If the two hash values are equal, the block is unaltered (not tampered with).
2. If the two hash values are not equal, the block has been tampered with.

Step 5: Bounding box is used to highlight the tampered region.

3.10 Recovery Process

After identifying the tampered regions of the image, the next phase will be the reconstruction of the image from the affected blocks. Image reconstruction is a vital part of the proposed technique because apart from detecting tampering, it helps in recovering the images that have been tampered with. Reconstruction can be achieved by using the recovery information that was embedded in the image. Even if there are modifications made in some sections of the image, the information provided by the recovery procedure will help in reconstructing those sections accurately.In the first phase of image reconstruction, the Most Significant Bits (MSBs) of the pixels that were tampered with will be restored. In every 8×8 pixel block, 32-bit representation of pixel data is developed, and the same is sent to the cloud (AWS). In the receiving section, these 32-bit representations corresponding to every block are extracted and used to restore the image.These bits include four bits per each pixel corresponding to its MSB value, which is assigned to the respective pixel in the block. Since the MSB contains the majority of the intensity information of the pixel, restoring these bits will lead to reconstruction of the image approximately to its original form.

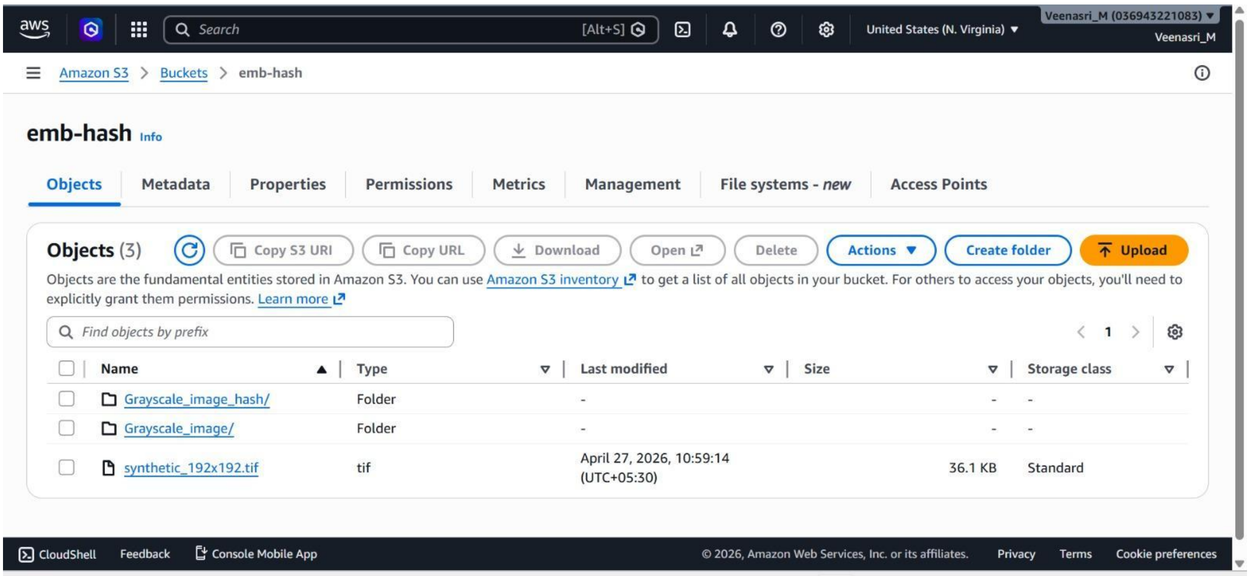


Fig 3.4 Amazon S3 cloud storage used in the proposed framework

Figure 3.4 illustrates the use of Amazon S3 cloud storage as a communication channel in the proposed tamper detection and recovery framework. The sender uploads the synthetic image along with its associated hash data to the cloud, which is then accessed by the receiver for integrity verification and recovery. This approach ensures reliable and efficient data transfer between system components.

3.11 Final Image Reconstruction

After recovering both the most significant bits (MSB) and least significant bits (LSB) of the tampered blocks, the final step in the process is image reconstruction. This step combines all the recovered information to generate the final output image.

The reconstruction process is carried out for each tampered 8×8 block. Once the MSB and LSB values are obtained, they are merged to form the complete 8-bit pixel value. This reconstructed pixel value is then placed back into its original position within the image.

Reconstruction Process

MSB Restoration

The first 4 bits of each pixel are restored using the transmitted pixel data.

LSB Restoration

The last 4 bits are recovered using the column-wise average values obtained from recovery data.

Bit Combination

The MSB and LSB are combined to form an 8-bit pixel value:

Pixel Value = (MSB<<4) + LSB

Block Replacement

The reconstructed pixel values are placed back into the tampered block.

Output of Reconstruction

The final image is obtained after processing all tampered blocks

The reconstructed image is an approximation of the original image

Some minor differences may exist due to data reduction, but overall quality remains high

CHAPTER-4

RESULTS AND DISCUSSIONS

4.1 Data Set and Metrics

All data generated or analysed during this study are included in this published article. All images are taken from the following web links https://sipi.usc.edu/database/, https://www.imageprocessingplace.com/root_files_V3/image_databases.html.

To evaluate visual quality, we use Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM),. Mean Squared Error (MSE) is used to measure the difference between two images, as shown in Equation (5), where I and K represent the two images, m and n represent the dimensions of the image, and i and j represent the coordinates of the pixel being processed. PSNR is calculated using Equation (6) and is inversely proportional to MSE. Therefore, the lower the MSE, that is, the higher the PSNR, the smaller the difference between the two images, indicating higher visual quality. SSIM is used to measure the similarity of two images, where x and y represent the two images, u represents the mean, o represents the standard deviation, and oxy represents the covariance between x and y, and C is a constant. The range of SSIM is from 0 to 1, and the closer the SSIM is to 1, the more similar the two images are.

$$\mathrm{MSE} = \frac{1}{p \times q} \sum_{i=1}^{p} \sum_{j=1}^{q} [\mathrm{I}\,(i,j) - \mathrm{K}\,(i,j)]^2$$

$$\mathrm{PSNR} = 10 \times log_{10}\ (255^2 / \mathrm{MSE})$$

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x \mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Variable Definitions:

- μx, μy: Local means of images x and y.
- σx^2, σy^2: Local variances of images x and y.
- σxy: Cross-covariance between x and y.
- C1, C2: Constants to stabilize division, defined as C1 = $(K1\ L)^2$ and C2 = $(K2\ L)^2$.
- L: Dynamic range of pixel values (e.g., 255 for 8-bit images).
- K1, K2: Small constants (default K1 = 0.01, K2 = 0.03).

The performance of the proposed tamper detection and restoration scheme was analyzed based on the PSNR, MSE, and SSIM criteria for various types of tampering and tampering ratios, as presented in Table 4.1. As can be seen from the results, the embedded and restored images preserve good visual quality and structural similarity compared to the original image. For instance, the restored image of the Airplane image under the Multiple Crop attack with a tampering ratio of 1.37% attains a PSNR value of 38.39 dB, an MSE value of 9.404, and an SSIM value of 0.985, signifying minimal distortion and maximum structural similarity upon reconstruction. Likewise, the restored image of the Barbara image under the Crop attack with a tampering ratio of 1.37% acquires a PSNR value of 40.1 dB and SSIM value of 0.99, implying successful restoration of the image.

These values demonstrate that the system maintains high image quality while providing reliable tamper detection and recovery.

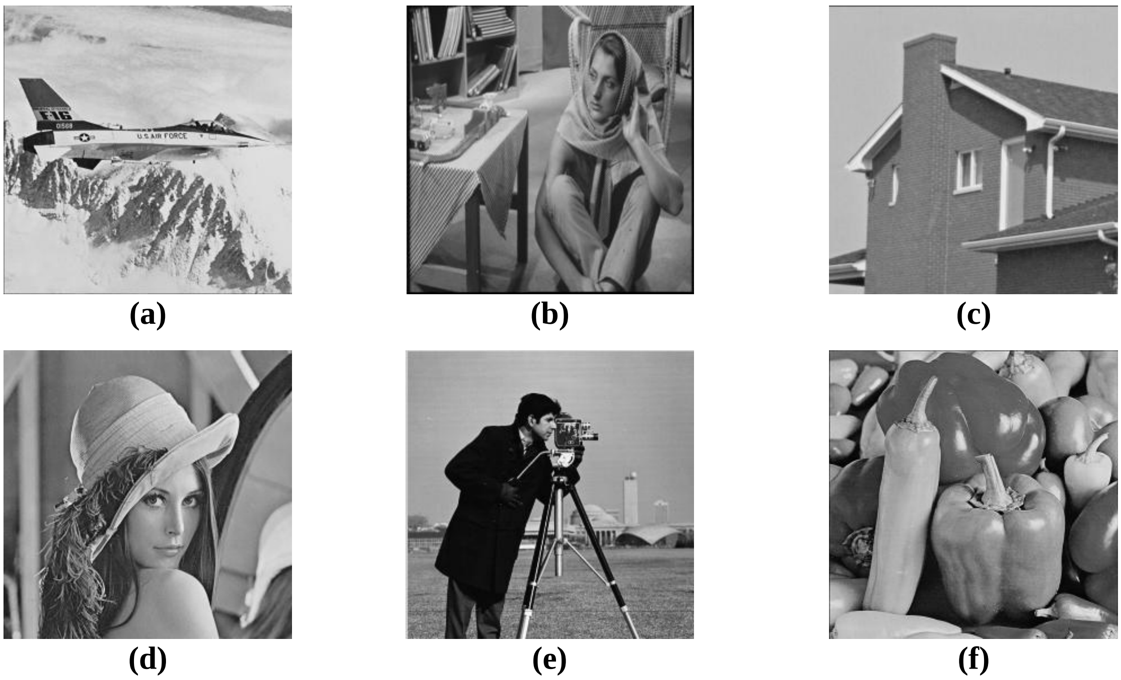
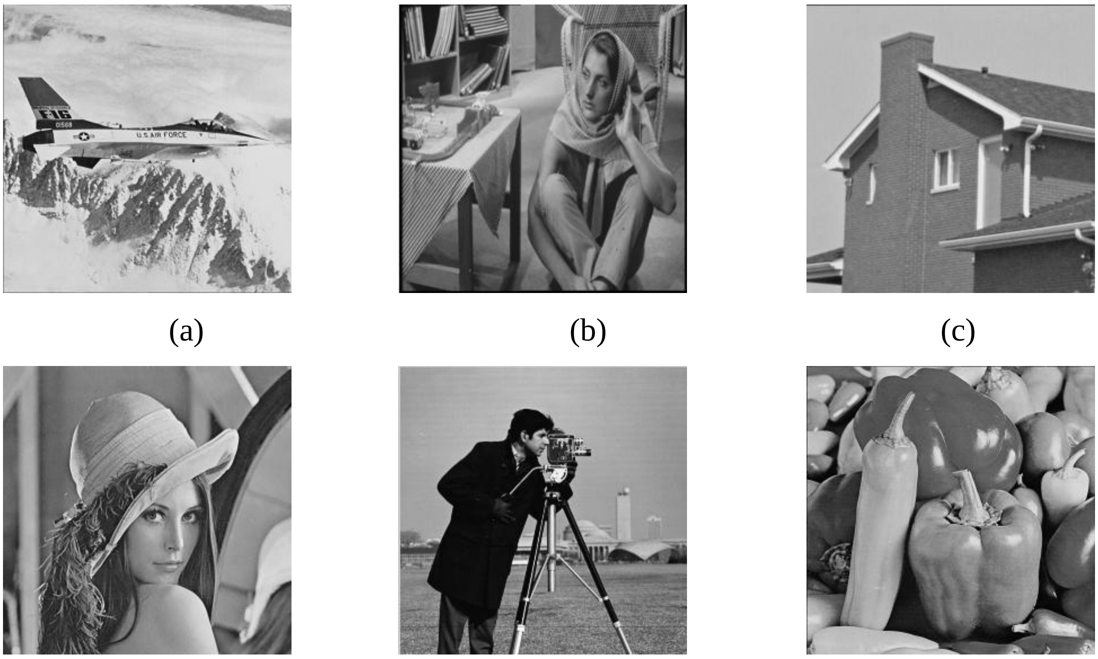


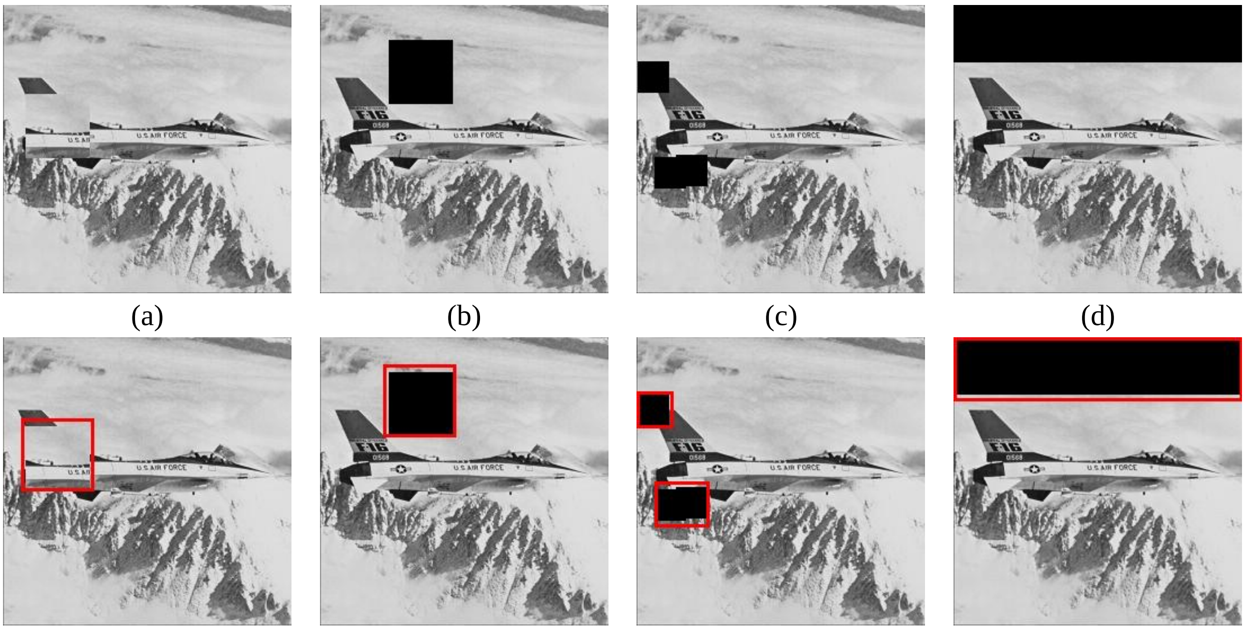
Fig 4.1 Original test images used for evaluation: (a) Airplane (b) Barbara (c) House (d) Lena (e) Cameraman and (f) Peppers.



(a) (b) (c)

(d) (e) (f)

Fig 4.2 Embedded Images of the original pictures: (a) airplane (b) barbara (c) house (d) lena (e) cameraman and (f) peppers



(a) (b) (c) (d)

(e) (f) (g) (h)

(i) (j) (k) (l)

Fig 4.3 Tampering and recovery results: (a) 5% copy-move attack, (b) 5% cropping attack, (c) multiple cropping attack (5%), and (d) top-strip removal attack (20%).(e)–(h) show the corresponding images for (a)–(d) with bounding boxes applied to highlight the tampered regions, respectively. (i)-(l) show the corresponding recovered images for (a)–(d), respectively.

Table 4.1

Performance analysis of tampering and recovery on Airplane, Barbara, pepper images using PSNR, MSE, and SSIM

Image			**Original vs Tampered**			**Original vs Recovered**		
	Attack name	**Attack percent**	**PSNR**	**MSE**	**SSIM**	**PSNR**	**MSE**	**SSIM**
Airplane	Multiple crop	1.37%	20.03	645.62	0.967	38.39	9.404	0.985
	Multiple crop	5%	18.95	826.6	0.9 5	26.56	143.2	0.95
	Multiple Crop	10%	12.71	3480.7	0.87	28.04	101.89	0.94
	Crop	1.37%	21.71	438.21	0.977	32.99	32.63	0.981
	Crop	5%	15.34	1900	0.934	37.7	10.8	0.97
	Crop	10%	13.46	2925.6	0.88	26.94	131.44	0.94
	Bottom-Strip	20%	9.59	7141.6	0.79	28.48	92.08	0.90
	Top-Strip	20%	9.4	7344.5	0.793	31.09	50.57	0.91
	Copy Paste	1.37%	31.3	47.44	0.98	33.19	31.17	0.984
	Copy Paste	10%	19.53	723.43	0.89	25.79	171.04	0.93
	Copy Paste	5%	20.92	525.09	0.946	27.85	106.6	0.96
Barbara	Multiple crop	1.37%	25.7	171.3	0.98	36.25	15.38	0.98
	Multiple crop	5%	18.15	993.9	0.93	34.01	25.8	0.97
	Multiple Crop	10%	16.6	1402.1	0.87	32.7	34.7	0.96
	Crop	1.37%	29.6	69.9	0.9	40.1	6.2	0.99
	Crop	5%	20.3	605	0.93	37.5	11.5	0.97
	Crop	10%	17.3	1207.3	0.89	34.1	24.8	0.96
	Bottom-Strip	20%	15.31	1912.5	0.79	28.78	86.05	0.92

	Top-Strip	20%	13.91	2640.5	0.798	27.82	107.26	0.921
	Copy Paste	1.37%	33.05	32.15	0.98	41.9	4.18	0.98
	Copy Paste	10%	19.8	671.2	0.9	32.1	39.5	0.94
	Copy Paste	5%	24.42	234.8	0.94	31.4	46.9	0.95
Pepper	Multiple crop	1.37%	25.4	184.09	0.97	36.14	15.79	0.98
	Multiple crop	5%	20.3	602.6	0.93	34.4	23.09	0.97
	Multiple Crop	10%	14.5	2265.2	0.87	30.14	62.89	0.94
	Crop	1.37%	22.9	333.4	0.97	40.8	5.3	0.99
	Crop	5%	17.7	1102.3	0.93	31.3	47.9	0.96
	Crop	10%	15.63	1775.5	0.88	30.5	57.5	0.95
	Bottom-Strip	20%	12.9	3320.5	0.79	26.81	135.3	0.90
	Top-Strip	20%	12.4	3689.6	0.79	27.69	110.54	0.91
	Copy Paste	1.37%	29.7	68.5	0.98	39.4	7.3	0.98
	Copy Paste	10%	21.08	506.9	0.9	32.9	33.1	0.95
	Copy Paste	5%	21.5	456.9	0.94	31.1	49.8	0.96

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

In this work, an image integrity model based on the usage of hashes for grayscale images has been successfully implemented in order to detect and recover tampered parts of a digital image.

The original image is split into blocks of 8×8 pixels, which are processed individually, thus providing different hash values and recovery data. The implementation of the SHA-256 algorithm allows for an efficient tampering detection procedure, owing to its highly sensitive nature, which responds to even the slightest modification of the pixel value. The inclusion of hash values and recovery data into the image makes the model self-contained, eliminating the necessity of having the original image available for the process.

At the receiving end, any tampering is identified by comparing the computed hash values with the extracted hash values. If there is any discrepancy, it means there has been tampering in the particular block. Tampering can be corrected by reconstructing the tampered blocks by restoring the MSBs from the transmitted data and recovering the LSBs using the recovery data.

From the experimentation carried out, it was found out that the proposed system offers excellent performance when considering the PSNR. It was found out that the embedded image had minimal distortions while the recovered image is similar to the original image.

In summary, the proposed system is efficient in identifying and correcting any tampering done on digital images.

5.2 Future Work

* Increase resistance to compression, noise, and geometrical attacks
* Increase effectiveness through adaptive or learning approaches
* Develop the approach for use on color images
* Ensure real-time efficiency

REFERENCES

1. Ozkaya, H. and Aslantas, V., 2024. A triple self-embedding fragile watermarking scheme for image tamper detection and recovery. *IEEE Access*, *12*, pp.140082-140096.

2. Vasu, S., George, S.N. and Deepthi, P.P., 2012, May. An integrity verification system for images using hashing and watermarking. In *2012 International Conference on Communication Systems and Network Technologies* (pp. 85-89). IEEE.

3. Gul, E. and Ozturk, S., 2019. A novel hash function based fragile watermarking method for image integrity. *Multimedia Tools and Applications*, *78*(13), pp.17701-17718.

4. Chaluvadi, S.B. and Prasad, M.V., 2009, December. Efficient image tamper detection and recovery technique using dual watermark. In *2009 World Congress on Nature & Biologically Inspired Computing (NaBIC)* (pp. 993-998). IEEE.

5. Renklier, A. and Öztürk, S., 2024. Image authentication and recovery: Sudoku puzzle and MD5 hash algorithm based self-embedding fragile image watermarking method. *Multimedia Tools and Applications*, *83*(5), pp.13929-13951.

6. Peñuelas-Angulo, A., Feregrino-Uribe, C., Cumplido, R. and Moreno-Ramírez, E., 2026. Reversible image authentication with efficient block-wise tamper localization. *Journal of Information Security and Applications*, *100*, p.104463.

7. Chaluvadi, S.B. and Prasad, M.V., 2009, December. Efficient image tamper detection and recovery technique using dual watermark. In *2009 World Congress on Nature & Biologically Inspired Computing (NaBIC)* (pp. 993-998). IEEE.

8. Hussan, M., Parah, S.A., Jan, A. and Qureshi, G.J., 2022. Hash-based image watermarking technique for tamper detection and localization. *Health and Technology*, *12*(2), pp.385-400.

9. Madhushree, B., Basanth Kumar, H.B. and Chennamma, H.R., 2024. An exhaustive review of authentication, tamper detection with localization and recovery techniques for medical images. *Multimedia Tools and Applications*, *83*(13), pp.39779-39821.

10. Su, G.D., Chang, C.C. and Chen, C.C., 2021. A hybrid-Sudoku based fragile watermarking scheme for image tampering detection. *Multimedia Tools and Applications*, *80*(8), pp.12881-12903.

Plagiarism Report

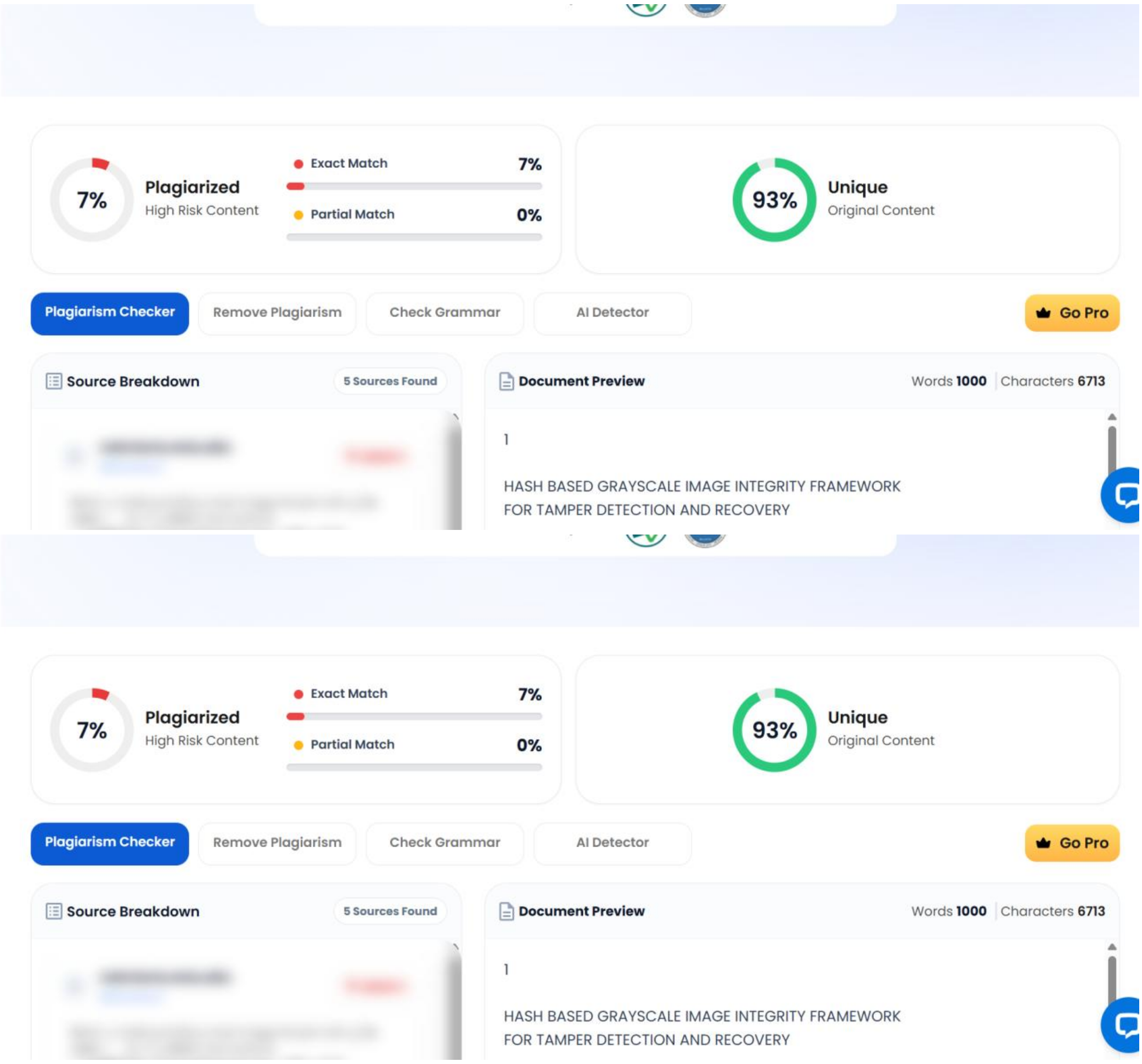


Fig 5 Plagiarism Report